

Bhargav Reddy Battu

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SUMMARY

Software Engineer with 2+ years of experience building scalable backend systems, AI-driven applications, and cloud-native infrastructure using Python, Java, FastAPI, Spring Boot, and AWS technologies. Developed distributed microservices, REST APIs, and analytics pipelines supporting payment processing, adaptive tutoring workflows, and multimodal inference systems across enterprise and educational environments. Experienced implementing Agentic AI orchestration, Retrieval-Augmented Generation (RAG), vector retrieval pipelines, and cloud deployment automation using PostgreSQL, Redis, Docker, and AWS services.

TECHNICAL SKILLS

Programming & Scripting: Python, Java, JavaScript, TypeScript, SQL, Bash, Shell Scripting

Frontend Development: Angular, React.js, Next.js, HTML5, CSS3, Tailwind CSS

Backend & API Development: Spring Boot, Spring MVC, Spring Security, Hibernate, FastAPI, Django, Flask, REST APIs, GraphQL, Microservices, Apache Kafka, Multithreading, HikariCP

AI & Machine Learning: Agentic AI, Retrieval-Augmented Generation (RAG), PyTorch, TensorFlow, LangChain, LangGraph, Claude AI API, MediaPipe, Vector Databases (Pinecone, FAISS, ChromaDB)

Cloud & DevOps: AWS Bedrock, AWS Agent Core Memory, AWS (EC2, Lambda, S3, RDS, IAM, CloudWatch, Amplify), Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins)

Databases & Tools: PostgreSQL, MySQL, MongoDB, Redis, Elasticsearch, Git, Linux, Data Aggregation

PROFESSIONAL EXPERIENCE

Software Engineer, AALMV

Jan 2026 - Present | FL, USA

- Engineered Python-based Agentic AI tutoring workflows using FastAPI, AWS Bedrock, and LangGraph, enabling adaptive reasoning across personalized AP Physics learning sessions daily for students.
- Optimized PostgreSQL semantic memory pipelines and Pinecone vector retrieval indexing supporting Retrieval-Augmented Generation workflows, improving contextual tutoring relevance by 32% during iterative student assessments sessions.
- Implemented FastAPI inference services integrating Claude AI APIs, Redis caching, and asynchronous orchestration pipelines supporting concurrent multimodal tutoring requests across scalable cloud learning environments daily.
- Constructed React and TypeScript educator dashboards with GraphQL progress services, enabling real-time mastery visualization and configurable recommendation tracking across instructional physics modules for instructors daily.
- Engineered highly available AWS infrastructure using EC2, IAM, CloudWatch, and GitHub Actions CI/CD pipelines, supporting reliable 24/7 tutoring platform operations across distributed environments for global students.
- Developed LangGraph-driven orchestration workflows integrated with Retrieval-Augmented Generation pipelines, Supabase PostgreSQL semantic storage, enabling reliable contextual reasoning across tutoring environments for students.

Software Engineer, University of Florida

Jun 2025 - Dec 2025 | FL, USA

- Integrated MediaPipe hand-tracking pipelines with Unity Animation Rigging constraints, improving dual-hand finger articulation stability by 30% during real-time gesture visualization experiments.
- Evaluated hand-tracking accuracy metrics through Python and .NET analytics workflows, improving experimental data reliability by 35% across iterative inference validation and testing sessions.
- Configured interactive Unity debugging interfaces supporting landmark freeze controls, bevel-angle visualization, and configurable UI states, increasing debugging efficiency by 25% during workflow evaluations.
- Automated multimodal tracking validation through CSV reporting and background inference monitoring pipelines, reducing manual verification workloads by 50% across gesture analysis experimentation environments.

Software Engineer, PhonePe

Aug 2022 - Dec 2023 | Remote, India

- Designed Java microservices architecture using Spring Boot, Spring MVC, and MySQL for centralized merchant transaction analytics across distributed payment processing environments.
- Analyzed transaction bottlenecks through Hibernate query profiling and aggregated reporting datasets, identifying synchronization inconsistencies affecting reconciliation workflows across regional payment services.
- Implemented Spring Security authentication, REST APIs, Apache Kafka event streams, and Java multithreading services, improving transaction reporting throughput by 28% during peak settlement operations.
- Validated AWS EC2-hosted microservices through Docker-based integration testing and scheduled monitoring pipelines, reducing backend synchronization failures by 24% across merchant support environments.
- Improved operational payment visibility by integrating HikariCP-optimized backend services with analytics dashboards, accelerating dispute investigation workflows by 31% for payment operations analysts.

CERTIFICATIONS

- Introduction to Embedded Systems Software and Development Environments - University of Colorado | Mar 2022
- Object-Oriented Data Structures in C++ - University of Illinois Urbana-Champaign | Coursera | Jun 2020 ([Link](#))
- Java Developer - JSpiders - Training & Development Center | Jul 2023
- Web Development - Teachnook (Teachscape Online Learning Services Private Limited - IIT Bhubaneswar) | Sep 2022

EDUCATION

Master of Science, University of Florida

Jan 2024 - Dec 2025 | FL, USA

Computer and Information Science

Bachelor of Technology, Vellore Institute of Technology

Aug 2019 - May 2023 | TN, India